

Week 2 Cardiorenal Protection Integration Guide

Primary Learning Objective

Students should learn to prioritize therapies that simultaneously:

- Reduce cardiovascular risk
- Slow CKD progression
- Reduce albuminuria
- Prevent future kidney failure
- Improve long-term patient outcomes

Management decisions should not be based on a single disease state in isolation.

Core Principle #1

CKD Is a Cardiovascular Disease Risk Multiplier

Patients with CKD are more likely to experience:

- Myocardial infarction
- Stroke
- Heart failure
- Cardiovascular death

Many CKD patients will experience a cardiovascular event before progressing to kidney failure.

Clinical implication:

Cardiovascular risk reduction should occur alongside kidney protection.

Core Principle #2

Albuminuria Matters as Much as eGFR

Students frequently focus on eGFR and overlook albuminuria.

Albuminuria is:

- A marker of kidney damage
- A marker of CKD progression risk

- A marker of cardiovascular risk

Always evaluate:

- eGFR
- UACR

Together rather than independently.

Core Principle #3

Therapy Selection Should Extend Beyond A1c

Historically, diabetes management focused primarily on glycemic control.

Modern management prioritizes:

- Kidney protection
- Cardiovascular protection
- Heart failure prevention
- Risk reduction

A patient can have:

- A1c at goal
- Progressive albuminuria
- Worsening CKD

and still require treatment intensification.

Cardiorenal Protection Framework

When evaluating a patient with CKD and cardiovascular risk, assess the following:

Step 1

Is the patient receiving appropriate lipid-lowering therapy?

Assess:

- ASCVD history
- Diabetes status
- CKD status
- LDL-C

Potential interventions:

- Initiate statin
- Intensify statin
- Add ezetimibe
- Consider PCSK9-directed therapy

Step 2

Is blood pressure optimized?

Assess:

- Current blood pressure
- Home blood pressure readings
- Medication adherence
- Albuminuria

Potential interventions:

- Optimize ACE inhibitor
- Optimize ARB
- Intensify antihypertensive therapy

Step 3

Is kidney-protective therapy optimized?

Assess:

- eGFR
- UACR
- Diabetes status

Potential interventions:

- Initiate SGLT2 inhibitor
- Optimize ACEi/ARB therapy
- Evaluate finerenone eligibility

Step 4

Is cardiovascular risk being addressed?

Assess:

- ASCVD history
- Smoking status
- Lipid control
- Blood pressure control
- Diabetes status

Potential interventions:

- Lifestyle modification
- Statin optimization
- Tobacco cessation
- Weight management

Medication Prioritization

ACE Inhibitor / ARB

Primary goals:

- Reduce albuminuria
- Slow CKD progression
- Improve cardiovascular outcomes

Monitor:

- Potassium
- Serum creatinine
- Blood pressure

SGLT2 Inhibitor

Primary goals:

- Slow CKD progression
- Reduce heart failure risk
- Reduce cardiovascular events

Monitor:

- eGFR
- Volume status
- Adverse effects

Finerenone

Primary goals:

- Reduce albuminuria
- Slow CKD progression
- Reduce cardiovascular risk

Monitor:

- Potassium
- Renal function

Statin Therapy

Primary goals:

- Reduce ASCVD risk
- Reduce cardiovascular events

Monitor:

- Lipid panel
- Adherence
- Tolerability

Clinical Decision-Making Scenarios

Scenario 1

Patient:

- T2DM
- eGFR 58 mL/min/1.73m²
- UACR 450 mg/g
- LDL-C 132 mg/dL

Clinical priorities:

1. Statin therapy optimization
2. ACEi/ARB optimization
3. SGLT2 inhibitor evaluation
4. Albuminuria reduction
5. Cardiovascular risk reduction

Scenario 2

Patient:

- LDL-C at goal
- Progressive albuminuria
- Worsening CKD

Clinical priorities:

1. Review ACEi/ARB therapy
2. Evaluate SGLT2 inhibitor use
3. Assess finerenone eligibility
4. Assess adherence
5. Intensify kidney-protective therapy

Do not assume management is optimized simply because LDL-C is controlled.

Scenario 3

Patient:

- A1c 6.8%
- eGFR declining
- Persistent albuminuria

Clinical priorities:

1. Kidney protection
2. CKD progression prevention
3. Albuminuria reduction

Do not focus exclusively on glycemic control.

Scenario 4

Patient:

- CKD
- Diabetes
- ASCVD history

Clinical priorities:

1. High-intensity statin therapy
2. ACEi/ARB optimization

3. SGLT2 inhibitor evaluation
4. Lifestyle modification
5. Ongoing monitoring

These patients frequently qualify for multiple evidence-based therapies simultaneously.

Monitoring Priorities

Assess routinely:

- eGFR
- UACR
- Blood pressure
- LDL-C
- Medication adherence
- Potassium
- Serum creatinine
- ASCVD risk factors

Trend data over time rather than relying on a single value.

Pharmacist-Specific Interventions

- Identify untreated albuminuria
- Optimize statin therapy
- Optimize ACEi/ARB therapy
- Identify SGLT2 inhibitor opportunities
- Evaluate finerenone eligibility
- Assess adherence barriers
- Review affordability concerns
- Perform renal dose adjustment review
- Develop monitoring plans
- Provide patient-centered counseling

Common Student Mistakes

- Treating CKD and ASCVD as separate disease states

- Focusing only on LDL-C
- Focusing only on eGFR
- Ignoring albuminuria
- Focusing only on A1c
- Missing opportunities for cardiorenal-protective therapy
- Not assessing medication adherence
- Delaying therapy intensification
- Failing to prioritize highest-risk problems first

Escalation Points

- Progressive CKD despite therapy
- Persistent severe albuminuria
- Rapid eGFR decline
- Recurrent hyperkalemia
- Refractory hypertension
- Established ASCVD with uncontrolled risk factors
- Failure to achieve risk-reduction goals
- Need for nephrology referral
- Need for advanced lipid-lowering therapy